Supplier databases

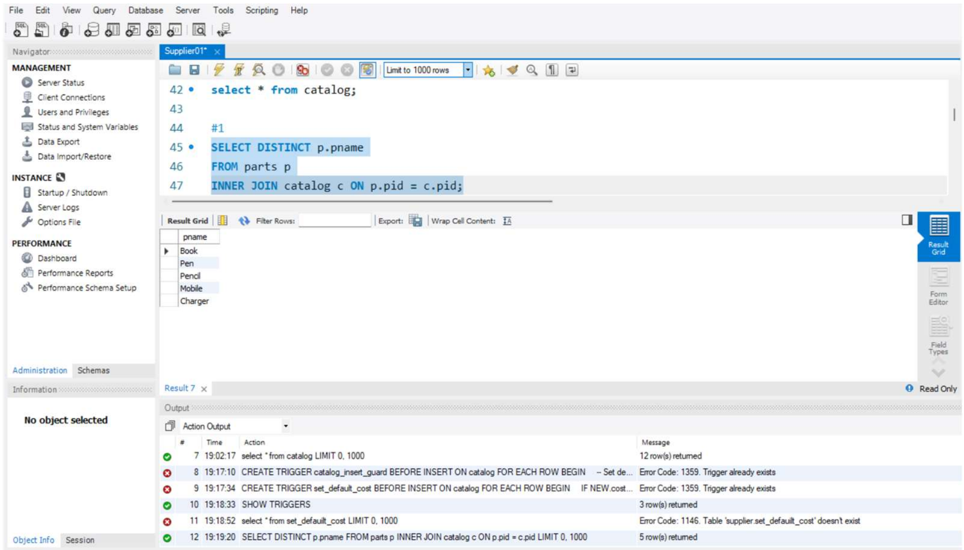
```
create database supplier;
use supplier;
 create table supplier(sid int primary key ,
 sname varchar(20) not  null,city varchar(20));

 create table parts( pid int primary key ,pname varchar(20)
 not null,color varchar(10));

 create table catalog(sid int ,pid int ,cost int ,
 foreign key(sid) references supplier(sid),
 foreign key(pid) references parts(pid));

 desc supplier;
 desc parts;
 desc catalog;

 insert into supplier values(10001,'Acme Widget','Bangalore');
insert into supplier values(10002,'Johns','Kolkata');
insert into supplier values(10003,'Vimal','Mumbai');
insert into supplier values(10004,'Reliance','Delhi');

select * from supplier;

insert into parts values(20001,'Book','Red');
insert into parts values(20002,'Pen','Red');
```

```
insert into parts values(20003,'Pencil','Green');

insert into parts values(20004,'Mobile','Green');

insert into parts values(20005,'Charger','Black');


select * from parts;


insert into catalog values(10001,20001,10);

insert into catalog values(10001,20002,10);

insert into catalog values(10001,20003,30);

insert into catalog values(10001,20004,10);

insert into catalog values(10001,20005,10);

insert into catalog values(10002,20001,10);

insert into catalog values(10002,20002,20);

insert into catalog values(10003,20003,30);

insert into catalog values(10004,20003,40);


select * from catalog;
```

Queries

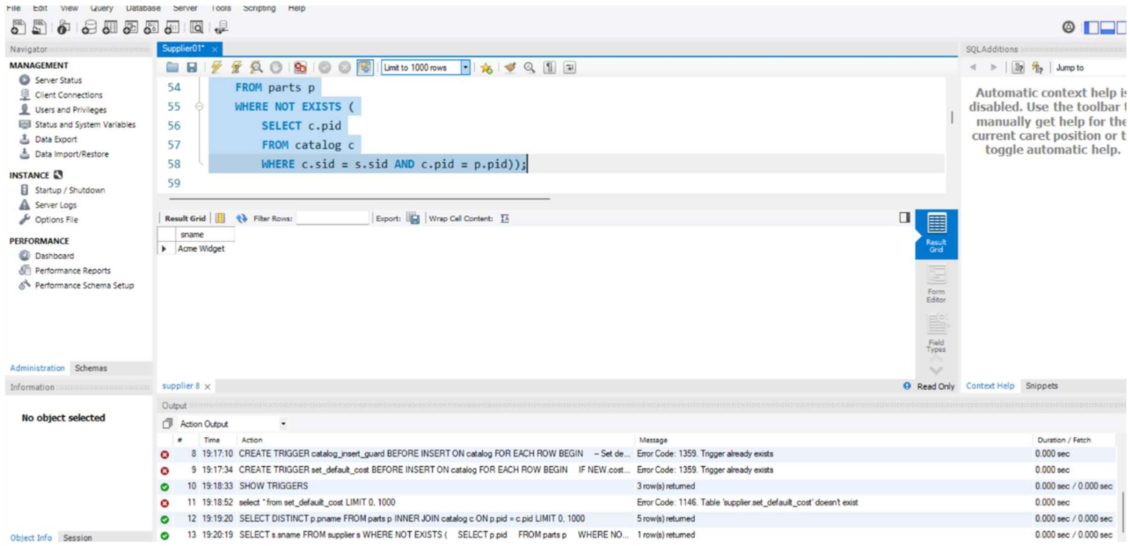
#1

```
SELECT DISTINCT p.pname

FROM parts p

INNER JOIN catalog c ON p.pid = c.pid;
```



#2

```
SELECT s.sname
FROM supplier s
WHERE NOT EXISTS (
   SELECT p.pid
   FROM parts p
   WHERE NOT EXISTS (
      SELECT c.pid
      FROM catalog c
      WHERE c.sid = s.sid AND c.pid = p.pid));
```



#3

SELECT p.pname

FROM parts p

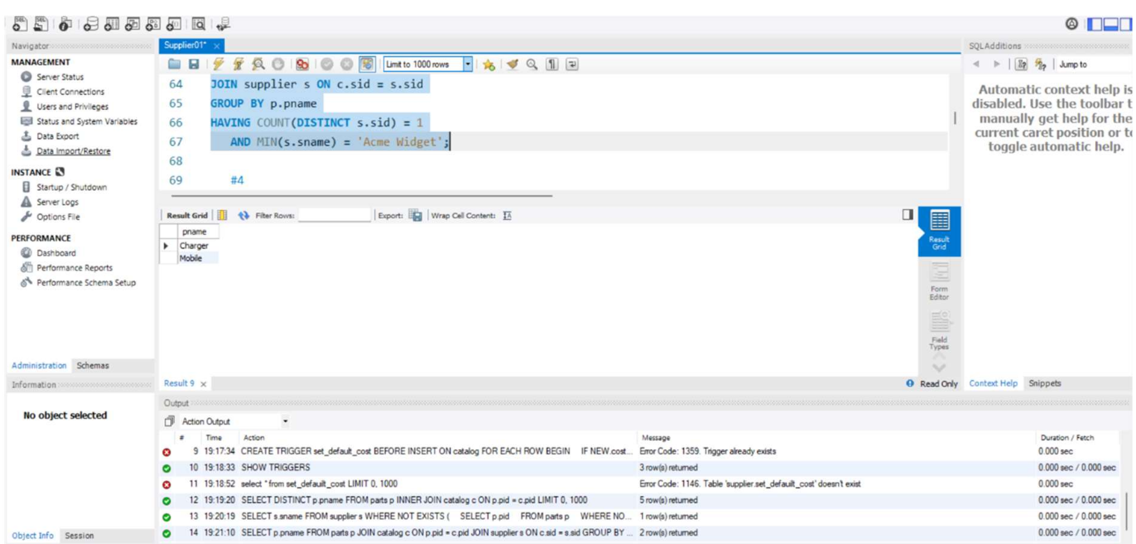
JOIN catalog c ON p.pid = c.pid

JOIN supplier s ON c.sid = s.sid

GROUP BY p.pname

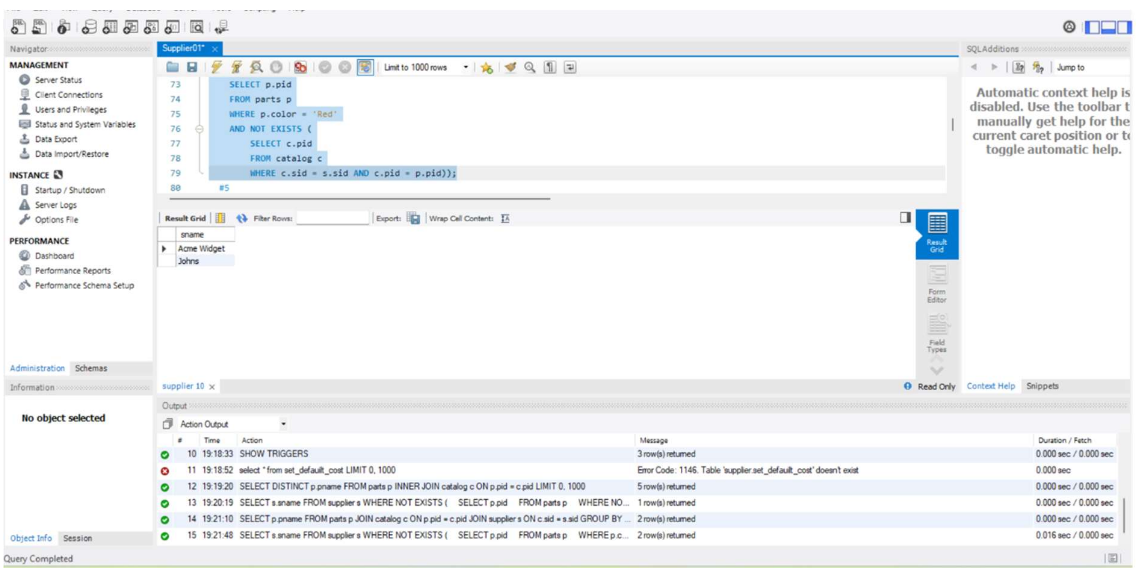
HAVING COUNT(DISTINCT s.sid) = 1

AND MIN(s.sname) = 'Acme Widget';



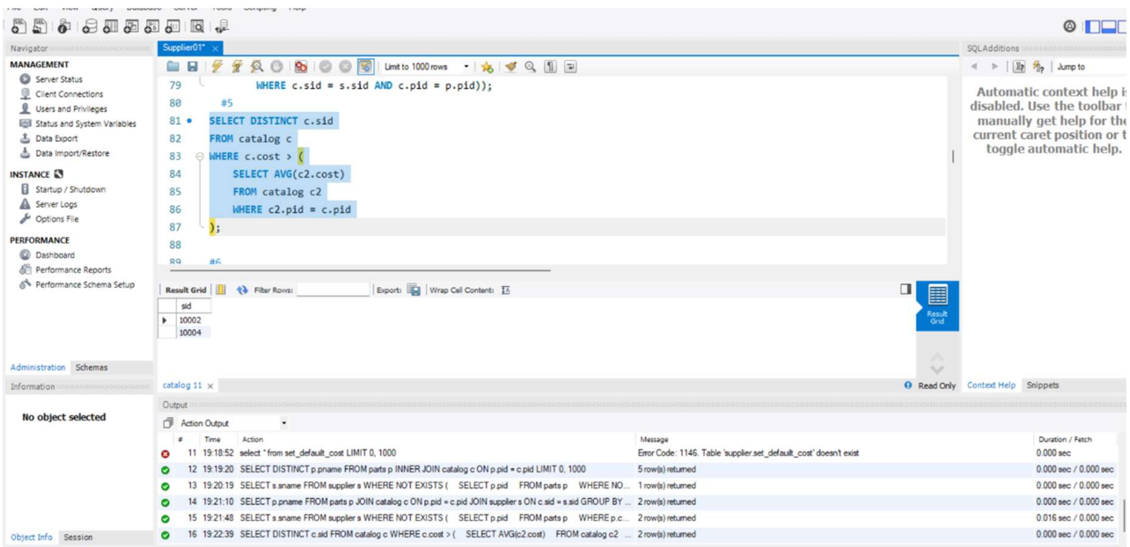
#4

```
SELECT s.sname
FROM supplier s
WHERE NOT EXISTS (
    SELECT p.pid
    FROM parts p
    WHERE p.color = 'Red'
    AND NOT EXISTS (
        SELECT c.pid
        FROM catalog c
        WHERE c.sid = s.sid AND c.pid = p.pid));
```



#5

```
SELECT DISTINCT c.sid
FROM catalog c
WHERE c.cost > (
   SELECT AVG(c2.cost)
   FROM catalog c2  WHERE c2.pid = c.pid);
```



#6

```
SELECT p.pname, s.sname, c.cost
FROM catalog c
JOIN supplier s ON c.sid = s.sid
JOIN parts p ON c.pid = p.pid
WHERE (c.pid, c.cost) IN (
   SELECT pid, MAX(cost)
   FROM catalog
   GROUP BY pid);
```